

CStats - Lab #2

March 3, 2026

Exercise 1.

Lake Macatawa, an inlet lake on the east side of Lake Michigan, is divided into an east basin and a west basin. To determine whether the bacteria count was lower in the west basin of Lake Macatawa than in the east basin, $n = 37$ samples of water were taken from the west basin and the number of bacteria colonies in 100 milliliters of water was counted. The sample characteristics were $\bar{x} = 11.95$ and $s = 11.80$, measured in hundreds of colonies. Find the approximate 95% confidence interval for the mean number of colonies (say, μ_w) in 100 milliliters of water in the west basin.

Solution 1.

$$\left[11.95 - 2.028 \left(\frac{11.80}{\sqrt{37}} \right), 11.95 + 2.028 \left(\frac{11.80}{\sqrt{37}} \right) \right] = [8.016, 15.884]$$

Exercise 2.

A light bulb manufacturer claims that the average lifespan of its bulbs is 170 hours. To verify this claim, a consumer protection agency randomly selects 100 bulbs from a production batch. After testing, they find an average lifespan of 158 hours with an empirical standard deviation of 30 hours. Assuming the lifespan follows a normal distribution, can we conclude from this survey that the manufacturer's claim is false?

Solution 2.

Hint. The manufacturer's claim can be validated by constructing a confidence interval that should include the value 170. By choosing a confidence level of 99% instead of the usual 95%, the obtained interval will be wider, which is more favorable to the manufacturer. To ensure the result cannot be disputed, a one-sided confidence interval of the form $m < \text{constant}$ can be chosen.

To construct a confidence interval for the mean m of this normal distribution with unknown standard deviation σ , we use the statistic:

$$\sqrt{n} \frac{\bar{X}_n - m}{S_n}$$

which follows a Student's t -distribution with $n - 1$ degrees of freedom. With $n = 100$, we find in the table the 0.995 fractile: $t = 2.63$. The symmetrical bilateral confidence interval is:

$$150 < m < 166.$$

The value indicated by the manufacturer is clearly outside this interval. If we choose a one-sided interval (to the left) at the same confidence level, it is defined as:

$$P\left(\sqrt{n}\frac{\bar{X}_n - m}{S_n} > t\right) = 0.99$$

where $t = 2.37$ and the interval is $m < 165$. Hence, there is a 99% chance that this advertisement is misleading.

Exercise 3.

A random sample of $n = 9$ wheels of cheese yielded the following weights in pounds, assumed to be $\mathcal{N}(\mu, \sigma^2)$:

21.50	18.95	18.55	19.40	19.15
22.35	22.90	22.20	23.10	

- (a) Give a point estimate of σ .
- (b) Find a 95% confidence interval for σ .
- (c) Find a 90% confidence interval for σ .

Solution 3.

For these 9 weights, $\bar{x} = 20.90$, $s = 1.858$.

- (a) A point estimate for σ is $s = 1.858$.

(b) $\left[\frac{1.858\sqrt{8}}{\sqrt{17.54}}, \frac{1.858\sqrt{8}}{\sqrt{2.180}} \right] = [1.255, 3.559]$

or

$$\left[\frac{1.858\sqrt{8}}{\sqrt{21.595}}, \frac{1.858\sqrt{8}}{\sqrt{2.623}} \right] = [1.131, 3.245];$$

(c) $\left[\frac{1.858\sqrt{8}}{\sqrt{15.51}}, \frac{1.858\sqrt{8}}{\sqrt{2.733}} \right] = [1.334, 3.179]$

or

$$\left[\frac{1.858\sqrt{8}}{\sqrt{19.110}}, \frac{1.858\sqrt{8}}{\sqrt{3.298}} \right] = [1.202, 2.894].$$

Exercise 4.

To verify the regularity of bus schedules, a bus company conducted 71 recordings of time gaps (in seconds) between bus arrivals at a stop and the observed average schedule. The unbiased estimate of the parameter σ^2 obtained was 5100. Deduce a 95% confidence interval for this measure of schedule regularity.

Solution 4.

Hint. We assume the classical hypothesis of measurement error normality, meaning that we must construct a confidence interval for the variance of a normal distribution, but this time with an unknown mean. Only the deviations from the empirical mean are considered here.

For a symmetrical bilateral confidence interval with 70 degrees of freedom the 0.025 and 0.975 fractiles are:

$$- a = 48.76$$

$$- b = 95.02$$

This gives the bilateral confidence interval:

$$3757 < \sigma^2 < 7322.$$

For a one-sided left confidence interval at the 0.05 fractile: $- a = 51.74$

The one-sided confidence interval is:

$$\sigma^2 < 6900.$$

Exercise 5.

To estimate the accuracy of a thermometer, $n = 100$ independent measurements are taken of a liquid maintained at a constant temperature of 20 degrees Celsius. The observations x_i , $1 \leq i \leq 100$, produced the value:

$$\sum_{i=1}^{100} x_i^2 = 40011,$$

Construct a 95% confidence interval for the precision of this thermometer, measured by the variance σ^2 of the measurements.

Solution 5.

Hint. Assuming the classical hypothesis of measurement error normality, we must construct a confidence interval for the variance of a normal distribution with a known mean $m = 20$. A symmetrical interval could be provided, but considering the parameter's interpretation, it is more logical to construct an interval giving an upper bound, which effectively measures the thermometer's imprecision.

We obtain the variance estimate:

$$\hat{\sigma}_{100}^2 = \frac{1}{100} \sum_{i=1}^{100} x_i^2 - m^2 = 0.11$$

From the χ^2 distribution table with 100 degrees of freedom the 0.025 and 0.975 fractiles are: - $a = 74.22$ - $b = 129.56$

This gives the symmetrical bilateral confidence interval:

$$0.08 < \sigma^2 < 0.15.$$

For a one-sided left interval (at the 0.05 fractile): - $a = 77.93$

The one-sided interval is:

$$\sigma^2 < 0.14.$$